BME 234 — Homework 2

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Problem 1: Logistic Regression Theory

(a) Prove equations (1) and (2) are equivalent.

Start with equation (1):

$$f(x) = \exp(\beta_0 + \beta_1 x) / [1 + \exp(\beta_0 + \beta_1 x)]$$

Divide both numerator and denominator by $\exp(\beta_0 + \beta_1 x)$:

$$= 1 / [1/\exp(\beta_0 + \beta_1 x) + 1]$$

$$= 1 / [1 + \exp(-\beta_0 - \beta_1 x)]$$

This is equation (2). QED.

(b) Prove that g′(z) = g(z)[1 – g(z)].

$g(z) = 1 / (1 + e^{-z})$

Using the quotient rule with $u = 1$, $v = 1 + e^{-z}$:

$$g'(z) = [0 \cdot (1 + e^{-z}) - 1 \cdot (-e^{-z})] / (1 + e^{-z})^2$$

$$= e^{-z} / (1 + e^{-z})^2$$

Factor:

$$= [1 / (1 + e^{-z})] \cdot [e^{-z} / (1 + e^{-z})]$$

$$= g(z) \cdot [(1 + e^{-z} - 1) / (1 + e^{-z})]$$

$$= g(z) \cdot [1 - g(z)] \quad \text{QED.}$$

(c) Rewrite the logistic regression model using g.

Let $z = \beta_0 + \beta_1 x$. Then:

$$f(x; \beta_0, \beta_1) = g(\beta_0 + \beta_1 x)$$

(d) Derive the log-likelihood.

Since $y_i \in \{0, 1\}$, the probability of observing y_i given x_i is:

$$P(y_i \mid x_i) = g(\beta_0 + \beta_1 x_i)^{y_i} \cdot [1 - g(\beta_0 + \beta_1 x_i)]^{(1 - y_i)}$$

This works because when $y_i = 1$ it gives g(...), and when $y_i = 0$ it gives 1 – g(...).

Since the data are i.i.d., the likelihood is the product over all observations:

$$L(\beta_0, \beta_1) = \prod_i g(\beta_0 + \beta_1 x_i)^{y_i} \cdot [1 - g(\beta_0 + \beta_1 x_i)]^{(1 - y_i)}$$

Taking the log converts the product to a sum:

$$\log L = \Sigma_i [y_i \cdot \log(g(\beta_0 + \beta_1 x_i)) + (1 - y_i) \cdot \log(1 - g(\beta_0 + \beta_1 x_i))]$$

QED.

(e) Why maximize log-likelihood instead of likelihood?

The logarithm is a monotonically increasing function, so any (β_0, β_1) that maximizes log L also maximizes L. The log transformation converts products into sums, making differentiation easier and computation more numerically stable (avoiding underflow from multiplying many small probabilities).

(f) Compute $\partial\log L/\partial\beta_0$ and $\partial\log L/\partial\beta_1$.

Let $z_i = \beta_0 + \beta_1 x_i$. Using the chain rule and $g'(z) = g(z)[1 - g(z)]$:

$$\partial\log L/\partial\beta_0 = \Sigma_i [y_i \cdot g'(z_i) / g(z_i) + (1 - y_i) \cdot (-g'(z_i)) / (1 - g(z_i))] \cdot 1$$

Substituting $g'(z_i) = g(z_i)(1 - g(z_i))$:

$$= \Sigma_i [y_i(1 - g(z_i)) - (1 - y_i)g(z_i)]$$

$$= \Sigma_i (y_i - g(z_i))$$

For β_1, $\partial z_i/\partial\beta_1 = x_i$, so:

$$\partial\log L/\partial\beta_1 = \Sigma_i (y_i - g(z_i)) \cdot x_i$$

(g) Gradient ascent update rules.

With learning rate α, repeat until convergence:

$$\beta_0 \leftarrow \beta_0 + \alpha \cdot \Sigma_i (y_i - g(\beta_0 + \beta_1 x_i))$$

$$\beta_1 \leftarrow \beta_1 + \alpha \cdot \Sigma_i (y_i - g(\beta_0 + \beta_1 x_i)) \cdot x_i$$

(h) Why might gradient ascent not converge with perfectly separable data?

If the data is perfectly linearly separable, the logistic regression can achieve zero loss by making the sigmoid arbitrarily steep — i.e., pushing $\beta_1 \to \infty$. The decision boundary becomes a step function. The gradients never reach zero because the log-likelihood keeps increasing (albeit more and more slowly) as the parameters grow. The parameters diverge to infinity, so gradient ascent never converges to finite, stable values.

(i) Fix for the convergence issue.

Add L2 regularization (ridge penalty). Instead of maximizing log L, maximize:

$$\log L - \lambda(\beta_0^2 + \beta_1^2)$$

The regularization term penalizes large parameter values, creating a counterforce that prevents the parameters from diverging to infinity. This ensures a finite optimum exists even when the data is perfectly separable. The hyperparameter λ controls the strength of the penalty.

Problem 2: Multiple Hypothesis Testing

(a) Bonferroni Correction

```
def apply_bonferroni_correction(pvalues, alpha):
    threshold = alpha / len(pvalues)
    return pvalues <= threshold
```

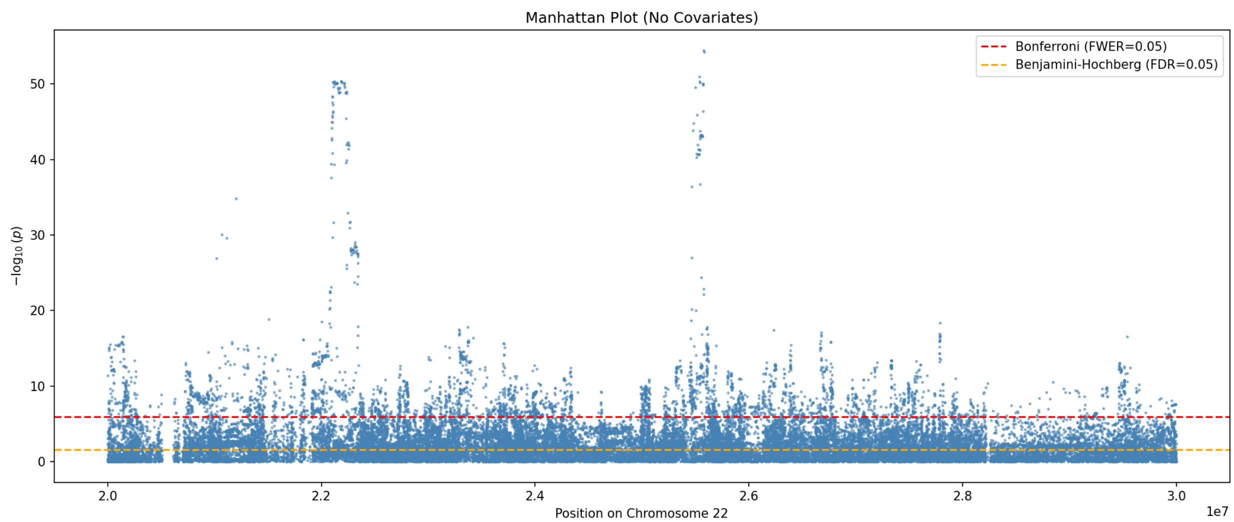
(b) Benjamini-Hochberg Correction

```
def apply_benjamini_hochberg_correction(pvalues, alpha):
    m = len(pvalues)
    sorted_indices = np.argsort(pvalues)
    sorted_pvals = pvalues[sorted_indices]

    thresholds = np.arange(1, m + 1) * alpha / m
    passing = sorted_pvals <= thresholds

    if not np.any(passing):
        return np.zeros(m, dtype=bool)

    max_k = np.max(np.where(passing))

    rejects = np.zeros(m, dtype=bool)
    rejects[sorted_indices[:max_k + 1]] = True
    return rejects
```

Problem 3: GWAS

(a)–(b) GWAS and Manhattan Plot (No Covariates)

See problem_3.py for the full implementation. The Manhattan plot below shows $-\log_{10}(p)$ for each SNP, with Bonferroni (red) and Benjamini-Hochberg (orange) threshold lines.



(c) Number of significant SNPs.

Without covariates:

- Bonferroni correction: 4,008 SNPs
- Benjamini-Hochberg correction: 24,389 SNPs

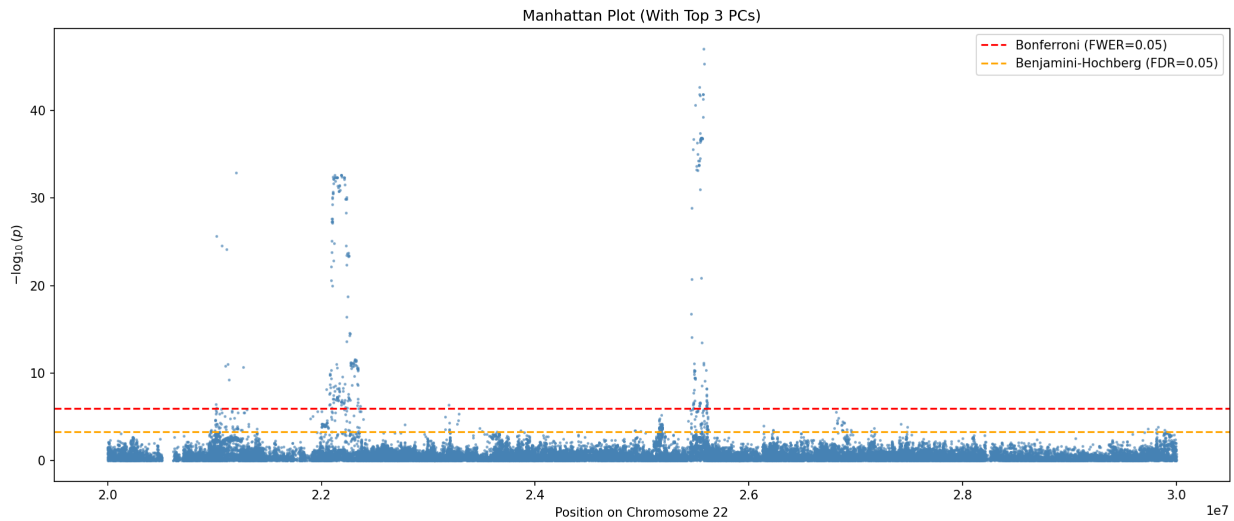
(d) Why so many significant SNPs when only 5 are causal?

1. Linkage disequilibrium (LD): SNPs that are physically close to causal variants on the chromosome tend to be inherited together. These nearby SNPs are correlated with the causal SNPs, so they also show statistically significant associations with the phenotype. This creates clusters of significant hits around each causal locus.

2. Population structure: The 1000 Genomes dataset contains individuals from diverse ancestral populations. Allele frequencies vary systematically between populations. If the phenotype prevalence also differs between populations, then any SNP whose frequency differs between populations will appear associated with the phenotype — even though the association is confounded by ancestry, not causality. This inflates the number of false positives across the entire chromosome.

(e) GWAS with Top 3 PCs as Covariates

The dosage matrix was standardized (mean 0, variance 1 per column) before PCA. The top 3 principal components were included as covariates in the logistic regression.



With PCA covariates:

- Bonferroni correction: 258 SNPs
- Benjamini-Hochberg correction: 520 SNPs

(f) Why does including PCs reduce significant SNPs?

The top principal components of the genotype matrix capture the major axes of genetic variation between individuals, which correspond to population structure (ancestry). By including them as covariates in the logistic regression, the model controls for ancestry differences. This removes spurious associations that arose because allele frequencies and phenotype prevalence both varied by population. After adjusting for population structure, only SNPs with true causal effects (and their LD neighbors) remain significant, dramatically reducing the number of hits and making it easier to identify the actual causal variants.